Deepak Mallampati

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Kent, OH | US-based, F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Applied AI engineer who ships production agent systems - schema-validated multi-agent pipelines (LangChain/LlamaIndex), RAG with ~35% retrieval precision, multi-LLM-provider integration (Claude/OpenAI/Ollama in Manga Lens), and FastAPI/Docker production services. Comfortable owning client engagements from concept through deployment.

SKILLS

Multi-Agent Workflows (LangChain/LlamaIndex) • Advanced RAG (Semantic Chunking, Embeddings) • LLM Provider Integration (OpenAI/Anthropic/Gemini) • Evaluation Pipeline Design • FastAPI / Docker Production • Client Engagement

Languages & Frameworks: Python, FastAPI, Flask, SQL (T-SQL, PostgreSQL, SQLite), TypeScript, React, C++ (Arduino), PHP

LLM & GenAI: LLM application development, Retrieval-Augmented Generation (RAG), agentic workflows, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, semantic retrieval, embeddings, vector search, responsible AI

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, Diffusers, LangChain, LlamaIndex

Computer Vision & Multimodal: YOLOv8, OpenCV, ControlNet, OpenPose/MediaPipe, Stable Diffusion, video analytics

Data & Analytics: Pandas, NumPy, feature engineering, time-series forecasting, walk-forward validation, NetworkX

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, RESTful APIs, cloud-ready deployment, observability/logging

Domains: Healthcare AI (HIPAA-conscious), Applied AI, Enterprise AI, Workflow Automation, Computer Vision, Multimodal Systems

PROFESSIONAL EXPERIENCE

Progress Solutions Inc. *Jul 2025 - Present*

Jr. AI/ML Engineer Trainee

USA - Healthcare Technology, AI Consulting, SaaS Healthcare

- Built **Retrieval-Augmented Generation (RAG) systems** for clinical knowledge retrieval and healthcare documentation search; implemented recursive semantic chunking and transformer-based embeddings; improved contextual retrieval precision by **~35%** and reduced irrelevant retrieval by **>30%**; retrieval-grounded response alignment exceeded **90%** in evaluation.
- Developed **agentic LLM workflows** for multi-step healthcare queries (eligibility checks, care workflow navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; improved agent response stability by **~25%**.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show prediction, care engagement scoring, and support prioritization; improved recall by **15-20%** on high-risk categories while holding precision **>90%** via class weighting, stratified sampling, and threshold calibration.
- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services, containerized with **Docker**, with structured logging and load simulation; reduced post-deployment defects by **~30%**.
- Built preprocessing pipelines (Pandas, NumPy) for EHR extracts, appointment histories, support ticket logs; raised dataset reliability above **98%** and cut downstream model instability by **~40%**.
- Maintained HIPAA-conscious data governance: de-identification, data lineage documentation, evaluation audit trails, and stakeholder-facing system-limitation docs.

Energy Solutions International (Emerson) *Jun 2022 - Apr 2023*

Database & DevOps Performance Engineer (Intern)

Hyderabad, India - Oil & Gas, Enterprise ERP, Industrial Automation

- Optimized SQL and **T-SQL stored procedures** powering the Synthesis Order-to-Cash platform (contracts, nominations, allocations, invoicing); reduced query execution time **~20%** and data retrieval latency **~25%**.
- Built **CI/CD pipelines with Jenkins** for schema updates, version-controlled SQL scripts, and stored procedure deployments; reduced deployment errors **>30%** and improved release cycle time **~35-40%** via structured release validation and rollback checkpoints.
- Designed performance dashboards using **SQL Server DMVs and Grafana** to track long-running queries, deadlocks, CPU/I/O contention; cut incident recurrence **~25%** and improved root-cause resolution speed **~30%**.
- Tuned database maintenance (index rebuilds, statistics updates, partition-aware indexing) and reconciliation queries; improved reconciliation job runtime **~15%** and reduced deadlocks **~15%**.
- Supported RBAC implementation and audit logging for financial modules in a compliance-sensitive oil & gas environment.

Suvidha Foundation *Jan 2022 - Mar 2022*

Machine Learning Engineer

Hyderabad, India - Nonprofit Tech, Video Analytics, Applied NLP

- Built a **transformer-based video summarization and highlight-generation system** across 5,000+ recorded sessions; transcript preprocessing, hierarchical summarization for long-context, and timestamp-aligned clip extraction reduced manual review time by **60-70%** (hours → <5 minutes per video).
- Achieved **~85% alignment** between AI-selected highlights and human-curated key moments.
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); reduced hallucinated outputs **~30%** and raised contextual answer accuracy **>85%** in controlled tests.
- Deployed the stack as a **Flask-based API** with a lightweight web interface for non-technical staff.

PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake Normalization → Validation → Consistency Analysis → Duplicate Detection → Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents; approximate-nearest-neighbor similarity search for duplicate claim detection; explainable risk scoring with audit-ready reasoning traces.

Manga Lens - Chrome Extension for Real-Time AI Manga/Webtoon Translation shipped, Chrome Web Store

Full-stack browser extension built independently - Manifest V3, content scripts, service workers, and `captureVisibleTab` for panel capture. Integrates **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v local). Multi-section capture pipeline handles tall webtoon panels via viewport-slice screenshots with coordinate remapping and dedup. Seven-day translation cache, per-domain selector configs for 29 manga/webtoon sites, privacy policy, and narrowed host permissions. Multi-provider payload handling (WebP for cloud, JPEG for Ollama to avoid CUDA crash).

Dream Decoder - AI-Powered Multimodal Creative Intelligence Platform

Full-stack FastAPI + React/TypeScript/Vite/Tailwind app that interprets dreams, generates poetic reinterpretations, and synthesizes dreamscapes via coordinated multimodal APIs (Perplexity Sonar for interpretation, Replicate image models for 16:9 visuals). Introduced **intermediate structured prompt transformation layers** - improved contextual alignment **~30%** and first-pass image success rate **~25-30%** over naïve direct-prompt orchestration.

Agentic Pixel Character Synthesis & Animation Engine ongoing

Phased generative AI system for identity-consistent, pose-controlled pixel character synthesis with sprite-sheet export. Stable Diffusion fine-tuning + LoRA for identity consistency, ControlNet + OpenPose/MediaPipe for pose conditioning, latent-space consistency for animation smoothness, and an **LLM-based agent orchestrator** that decomposes high-level prompts into generation tasks. Backend: PyTorch, Hugging Face Diffusers, FastAPI, Docker.

EDUCATION

Master's Degree - *Kent State University, Ohio, USA* *Aug 2023 - May 2025*